**ML-Based Attendance Certification System By Manish Dhandapani(24781A33A1)-SVCET**

## 1. Introduction

Attendance management is an essential part of academic and internship programs. Manually verifying attendance records for a large number of participants can be time-consuming and error-prone. This project aims to automate the attendance verification and certification process using data preprocessing, analytics, and machine learning techniques.

The system processes attendance records of 980 students across 40 internship sessions and determines certification eligibility based on predefined attendance criteria. The project demonstrates the complete data science workflow, including data cleaning, feature engineering, exploratory data analysis (EDA), machine learning model training, and report generation

## 2. Problem Statement

An internship program consists of 40 training sessions, each having a duration of 120 minutes. A student is considered present for a session only if their attendance duration is at least 90 minutes.

Certification Criteria:

- Total Sessions: 40
- Session Duration: 120 Minutes
- Minimum Attendance per Session: 90 Minutes
- Minimum Required Attendance: 80%
- Minimum Sessions Required for Certification: 32

The objective is to identify students who satisfy the certification criteria and generate a final attendance report containing certification status.

## 3. Dataset Description

A synthetic attendance dataset was created to simulate real-world internship attendance records.

### Student Master Sheet

Contains information about all students:

- Student_ID
- Student_Name
- Email

Total Students: 980

### Session Sheets

The dataset contains 40 attendance sheets:

- Session_01

- Session_02
- ...
- Session_40

Each sheet contains:

- Student_ID
- Student_Name
- Email
- Attendance_Minutes

**Realistic Data Challenges Introduced**

To simulate real attendance records, the dataset includes:

- Duplicate entries
- Missing email values
- Inconsistent name formatting
- Attendance outliers
- Irregular attendance patterns

These issues were intentionally introduced to perform meaningful preprocessing and data cleaning.

**4. Data Preprocessing**

Data preprocessing was performed before analysis and model training.

**Cleaning Operations**

1. Duplicate Removal
    - Removed repeated attendance records.
2. Missing Value Handling
    - Missing email values were filled using Student_Master data.
3. Name Standardization
    - Student names were converted into a consistent format.
4. Outlier Treatment
    - Attendance values exceeding the maximum session duration were corrected.
5. Attendance Flag Generation
    - Present = 1 if Attendance_Minutes ≥ 90
    - Present = 0 otherwise

The cleaned data was then used for feature engineering and model development.

**5. Feature Engineering**

After preprocessing, student-level features were generated from the attendance records.

**Features Created**

- Sessions_Attended
- Attendance_Rate
- Missed_Sessions
- Consistency

**Target Variable**

Certification status was generated using the project criteria.

Certified = 1 if Sessions_Attended ≥ 32

Certified = 0 otherwise

These features were used as inputs for machine learning models.

**6. Exploratory Data Analysis (EDA)**

EDA was performed to understand attendance patterns and student participation behavior.

**Analysis Performed**

- Attendance distribution analysis
- Certification statistics
- Attendance rate analysis
- Correlation analysis
- Attendance pattern visualization

**Key Observations**

- Highly active students attended most sessions consistently.
- Irregular students showed significant variation in attendance.
- Attendance rate strongly influenced certification eligibility.
- Most certification decisions could be explained through attendance-related features.

**7. Machine Learning Models**

The problem was treated as a binary classification task where the objective was to predict whether a student would be certified or not.

**Models Implemented**

**Logistic Regression**

A linear classification algorithm used as a baseline model.

**Decision Tree Classifier**

A tree-based model that creates decision rules based on attendance features.

**Random Forest Classifier**

An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

**Evaluation Metrics**

The models were evaluated using:

- Accuracy
- Confusion Matrix
- Classification Report

Among the implemented models, Random Forest produced the best overall performance due to its ability to capture complex attendance patterns.

**8. Results and Output**

The system successfully generated certification results for all students.

**Final Output**

The project generates:

**ML_Attendance_Report.xlsx**

The workbook contains:

- Student_Master
- Session_01 to Session_40
- Final_Attendance_Summary

**Final Attendance Summary**

The summary sheet includes:

- Student_ID
- Student_Name
- Email
- Sessions_Attended
- Attendance_Rate
- Certification_Status

This enables both administrators and students to easily verify certification eligibility.

**9. Technologies Used**

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- OpenPyXL
- Jupyter Notebook

**10. Conclusion**

The ML-Based Attendance Certification System successfully automates attendance analysis and certification generation for large-scale internship programs. The project demonstrates practical applications of data preprocessing, exploratory data analysis, feature engineering, and machine learning in solving real-world administrative problems.

The final system provides an efficient and scalable solution for identifying eligible students while maintaining transparency in the certification process.

**Future Enhancements**

- Real-time attendance tracking
- Web-based dashboard
- Automated certificate generation
- Predictive attendance analytics
- Cloud deployment and reporting